

Lab 5: PageRank

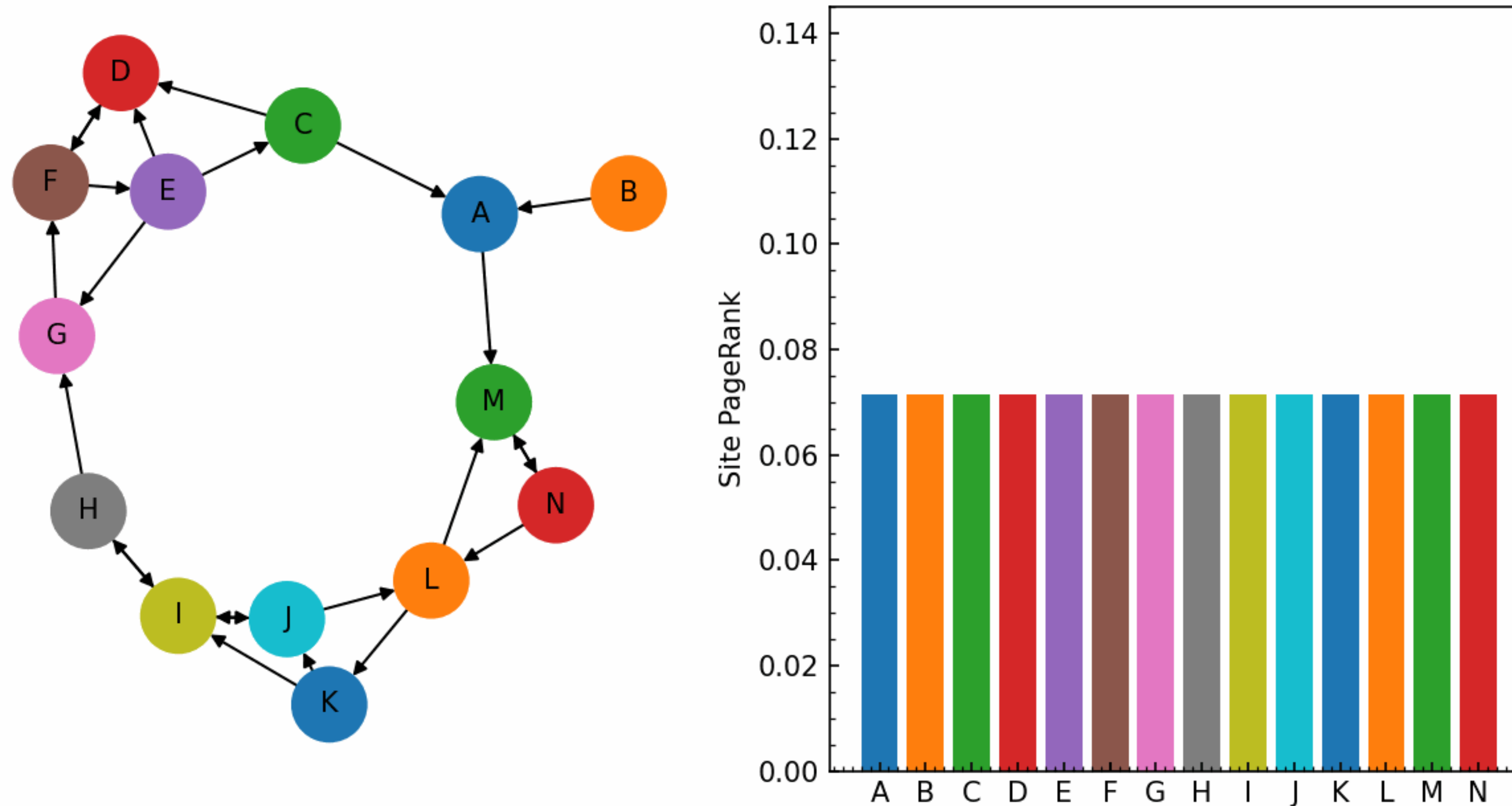
CAS CS 132: Geometric Algorithms

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April 6th, 2026

What is PageRank?

The *first* algorithm used by Google Search to rank web pages



https://en.wikipedia.org/wiki/PageRank#/media/File:Page_rank_animation.gif

Research paper from Google: <https://research.google/pubs/the-anatomy-of-a-large-scale-hypertextual-web-search-engine/>

Markov chains & steady states

A quick review

- A Markov chain describes transitions between states using probabilities.
- The steady state is a vector s such that $As = s$: once you reach it, the distribution does not change anymore.
- In PageRank, that steady-state distribution tells us how important each webpage is.

Setup

Represent the web (1/2)

- Represent the world wide web as a *directed graph*, where
 - Node u : a webpage
 - Directed edge $e = (u, v)$: a link of webpage v , present in webpage u
- A random user navigates the web, i.e., keeps clicking links.

Setup

Represent the web (2/2)

Observe that A is sparse (why?)

- Let the stochastic matrix $A \in \mathbb{R}^{n \times n}$ where,
 - column i corresponds to page i
 - if page i links to page j , then $A_{j,i} \neq 0$
 - if page i has k outgoing links, then each linked page gets probability $1/k$
 - $\Rightarrow$ each nonzero column sums to 1

Setup

Represent a user

- Let column vector $u = [1/n, \dots, 1/n]^T \in \mathbb{R}^n$
- The surfer i starts at a (uniformly) random webpage
- Then:
 - after one click, distribution is Au
 - after k clicks, it is $A^k u$
- As $k \rightarrow \infty$, this approaches the **steady-state distribution of A** : $\lim_{k \rightarrow \infty} A^k u$
- That limit is the **PageRank vector**; and the **eigenvector for eigenvalue 1!**

2.1.2. Intuitive justification

PageRank can be thought of as a model of user behavior. We assume there is a “random surfer” who is given a Web page at random and keeps clicking on links, never hitting “back” but eventually gets bored and starts on another random page. The probability that the random surfer visits a page is its PageRank.

Why not solve for the eigenvector directly?

The PageRank vector is also the eigenvector of eigenvalue 1

- In theory, we could.
- In practice, the web graph is enormous, so:
 - the matrix A is huge
 - exact eigendecomposition would be far too expensive
- But also, the matrix A is sparse
- And it happens that $A^k u$ is a good approximation for relatively small k
- Hence, we use the **power method: repeatedly multiply by the matrix.**

Power method

- An iterative algorithm for approximating the steady-state vector by repeatedly applying the transition matrix to an initial guess.
- $v^{(0)} = \frac{1}{n}\mathbf{1}$
- For $t = 0, 1, 2, \dots$:
- $v^{(t+1)} = \text{power_step}(v^{(t)})$
- Stop when $\|v^{(t+1)} - v^{(t)}\|_1 < \varepsilon$

Let's fix some issues

Boundaries/dangling nodes

2.1.2. Intuitive justification

PageRank can be thought of as a model of user behavior. We assume there is a “random surfer” who is given a Web page at random and keeps clicking on links, never hitting “back” but eventually gets bored and starts on another random page. The probability

- **Issue:** Some webpages may have no outgoing links.
 - Then, its column in matrix A will be all zeros
 - And if the user lands there, it's a dead-end.
- **Fix:** let's transform all zero columns to all-ones columns (before normalization).
 - Let the user navigate to any webpage from that dangling page.

Let's fix some issues

Teleportation

2.1.2. Intuitive justification

PageRank can be thought of as a model of user behavior. We assume there is a “random surfer” who is given a Web page at random and keeps clicking on links, never hitting “back” but eventually gets bored and starts on another random page. The probability

- **Issue:** the user may get trapped in cycles of connected components of the graph.
- **Fix:** use **teleportation**. w.p. $1 - \alpha$, follow A
 - Transition using $(1 - \alpha)A + \frac{\alpha}{n}\mathbf{1}$ w.p. α , teleport
- The PageRank paper uses $\alpha = 0.15$

Wait, is this $(1 - \alpha)A + \frac{\alpha}{n}\mathbf{1}$ sparse anymore?

Every column is nonzero now; the matrix is now dense

- Let's rewrite things...

- $A = \boxed{A} + \boxed{\frac{1}{n}A_z}$ A is the initial sparse matrix (possibly with zero columns)
 A_z has all-ones columns exactly where A had zero columns,
and zeros elsewhere

- $\left((1 - \alpha)A + \frac{\alpha}{n}\mathbf{1} \right) u = \boxed{(1 - \alpha)Au + \frac{1 - \alpha}{n}A_z u + \frac{\alpha}{n}\mathbf{1}u}$

Transition formula/power step

Convergence check

Using the L1 error

- If the current iteration is v_t and the previous one is v_{t-1} , let
- $$\text{error}_{\ell_1}(v^{(t-1)}, v^{(t)}) = \sum_{i=1}^n |v_i^{(t-1)} - v_i^{(t)}|.$$
- When error < tolerance, stop the power iteration.

- **Link to Lab 5:** <https://nmmull.github.io/cs132-s26/labs/5/main.pdf>
- **Reading:** <https://mcrovella.github.io/CS132-Geometric-Algorithms/L19PageRank.html#pagerank>
- **Write-up due:** April 16, 2026 by 8:00PM